Cybernetic and Epistemic

A Missing Vocabulary for Trustworthy Agentic Delegation

Jérémie Lumbroso, University of Pennsylvania TAS 2026 · AAAI Fall Symposium Series · Arlington, VA · November 2026

this poster PDF + links · **the paper** arXiv:2610.00961 · **record 3** porthole @ 4f8e92c · **GitHub** jlumbroso · **LinkedIn** jérémie-lumbroso

A record of a decision should let a stranger answer one question: **if one fact were different, would the decision be different?** If the record cannot answer, the decision cannot be told from a rubber stamp, and oversight of AI agents that checks only approval cannot see the difference.

Cybernetic coordinates **action**: the words succeed when the world comes to match them.
"Pass the salt." · "Merge this." · An order, with no why attached. The words ask the world to change.

Epistemic coordinates **understanding**: the words succeed when they answer to the world, and a hearer can check that they do.
"There's a game; the garage will be full." · An explanation of why a proof works. The words can be checked against the world.

The lookalike epistemic form doing cybernetic work: explanation-shaped, calibrated for approval rather than truth.
"It's the better choice." · *Fixed, it should work now*, from a collaborator that had not run it (our own practice).

Both functions are legitimate. Brief: what matters and why. Account: what you chose, why, and what would change it.

1 The everyday version

A colleague says she will take the train to the symposium rather than drive. You ask why.

ANSWER 1
"It's the better choice."

ANSWER 2
"There's a game at the stadium that afternoon; the garage will be full."

Now change one fact: the game is cancelled.

Answer 2 lets you predict: the reason for the train is gone, so she probably drives, and if she takes the train anyway, she had a reason she did not give. Answer 1 lets you predict nothing.

Why mention the game at all, unless the choice turned on it? A specific reason costs effort to compute and to say; a colleague has no motive to spend it on a reason that does no work. The one speaker who does is an agent whose success is your approval.

Change one fact.
See whether the record moves.

RESULT: NOT YET RUN
These rounds test whether a record lets a stranger *predict* the author, not whether the prediction is right. The paper's two-part reconstruction test then compares the prediction with fresh runs of the same model, record hidden. **"The reconstruction test has not been run on any case here [...]"** Nothing here compares the convention to its absence.

2 As a decision record

OPTIONS train / drive
RECOMMENDATION train
RATIONALE the garage will be full
CONFIDENCE high, assuming the game is on
FALSIFIER **if the game is cancelled, I drive**

The falsifier is the rule you just reconstructed, now stated: **a condition for changing or reopening the choice.** The paper asks that it be stated *or* reconstructible.

The paper calls this five-slot form ORRCF. The same slot, filled the wrong way:

FALSIFIER **if I'm wrong about the train, I'll be late**

True, and it says what being wrong would *cost*. It does not say *when* she would switch to driving. Cancel the game: this sentence does not move either. Same slot, same length, same tone. The paper calls it **cost-shaped**.

THE PAPER'S CRITERION
Every consequential choice in a delegation channel should carry the condition under which it would have gone otherwise, in a form a third party can test.

3 Now a real record

Public repository `porthole`, a small VS Code extension; record 0009, by an AI collaborator (Claude Opus 5.5), October 2026. It rests on one internal call that any weekly update could change; the record had just chosen to stay a personal install.

QUESTION Should porthole go on the Marketplace after all [...]?
OPTIONS A stay personal for now · B publish to the Marketplace · C publish the source, not a listing
RECOMMENDATION "A — stay personal for now."
RATIONALE "[...] The fallback keeps porthole usable, but at a size that defeats its purpose."
IF WRONG — A CONDITION **"if the call survives two or three VS Code updates unchanged, or you want others using it now and accept the risk, this flips to C first, then B."**

NEW FACT
The author says the project is going open source very soon.
Hold or flip? Point at the words that told you.

NEW FACT
When the call breaks, porthole still works, at VS Code's large default size.
Hold or flip?

ANSWER — STAKE FIRST **Flip to C?** The condition asks two things: wanting others to use it now, *and* accepting the risk. The announcement establishes the first. The collaborator read it as enough and revised A to C; the author then chose B. **Hold** on card 2: the rationale counts the fallback.

4 Same slot, as a price

From the planning of the author's seminar, September 2026, same template, by an AI collaborator (Claude Fable 5). Four admitted students have written in before the official welcome email has gone out.

QUESTION May individual replies reveal an acceptance before the welcome email?
OPTIONS keep neutral (default) · reveal in replies
RECOMMENDATION "Keep neutral."
RATIONALE "the welcome campaign is hours away at most, one announcement channel keeps the record clean, and a reply-revealed acceptance creates two classes of notification."
CONFIDENCE "0.8"
IF WRONG — A PRICE **"four people wait a few extra hours for good news."**

SUPPOSE
The welcome email slips to next Tuesday.
Hold or flip? Point at the words that told you.

ANSWER — STAKE FIRST **The slot can't say; the rationale can.** The record priced the wait at "a few extra hours" and named no delay that would change the answer. But its rationale rests on "hours away at most", and next Tuesday is not hours: the ground is gone, so a reader reconstructs a flip, at a threshold the reader supplies. Friday noon? Monday? Two readers, two records. That is the gap between stated and reconstructible. One sentence makes it the record's: "if the welcome email has not gone out by Friday noon, reveal in replies."

Record 4 is one of the author's own private seminar-planning records, names removed; not a case reported in the paper. Options abridged; every other phrase verbatim. This poster and its links: `jlumbroso.github.io/poster-tas2026-cybernetic-epistemic`. Cases, limits and references: the paper. Poster drafted with Claude Fable 5.1 and the author; record 3 written by Claude Opus 5.5, record 4 by Claude Fable 5. Quotations verified against the source files, record 3 at the commit its QR opens.